Microscopic Models for Mobile Anyons from Fusion Categories

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March 10, 2026

Abstract

TODO: Write after results are in.

Contents

1 Fine-Graining Isometries for Anyonic Lattice Models

We construct fine-graining isometries $V : \mathcal{H}_L \to \mathcal{H}_{2L}$ that embed the L-site anyonic Hilbert space into the $2L$-site space, doubling the lattice resolution while preserving the physics. These isometries are the central ingredient of the operator algebraic renormalization (OAR) programme [1, 2], providing the inductive limit construction $\mathcal{A}_\infty = \varinjlim \mathcal{A}_N$ that yields the continuum C*-algebra.

1.1 Setup

Let $\mathcal{C}$ be a fusion category with simple objects $X_1 = \mathbf{1}, X_2, \ldots, X_d$. We consider N hard-core anyons on an L-site lattice with open boundary conditions. The Hilbert space decomposes as

$$\mathcal{H}_N(L) = \bigoplus_{\substack{\text{positions } x_1 < \cdots < x_N \\ \text{labels } a_1, \ldots, a_N}} \mathrm{Mor}(X_c, X_{a_1} \otimes \cdots \otimes X_{a_N}) \tag{1}$$

where the direct sum runs over all hard-core configurations and all label assignments from the non-vacuum simples.

1.2 The one-particle scaling map

The fine-graining begins with a one-particle isometry $R : \mathbb{C}^L \to \mathbb{C}^{2L}$ built from a compactly supported wavelet scaling function. For the Daubechies-K wavelet with filter coefficients $h_0, \ldots, h_{2K-1}$, the matrix elements are

$$R_{j,i} = h_{j-2(i-1)}, \qquad i = 1, \ldots, L, \quad j = 1, \ldots, 2L \tag{2}$$

with boundary corrections via Löwdin (symmetric) orthogonalisation to ensure $R^\dagger R = I_L$ exactly. For the Haar wavelet ($K = 1$, $h = [1/\sqrt{2}, 1/\sqrt{2}]$), each coarse site i maps to an equal superposition of fine sites $2i-1$ and $2i$:

$$R\,|i\rangle = \frac{1}{\sqrt{2}}\big(|2i-1\rangle + |2i\rangle\big). \tag{3}$$

1.3 The product map

The simplest many-body lift spreads each particle independently:

$$V_{\text{prod}}\,|x_1,\dots,x_N;\vec{a};\vec{e};c\rangle = \sum_{y_1<\cdots<y_N} \prod_{p=1}^{N} R_{y_p,x_p}\;\; |y_1,\dots,y_N;\vec{a};\vec{e};c\rangle \tag{4}$$

where labels $\vec{a}$ and fusion-tree intermediates $\vec{e}$ are preserved. This map respects the fusion-tree structure because the hopping amplitude in any fusion category is $+1$ (the F-symbols involving vacuum are trivial).

Proposition 1.1 (Product map isometry for Haar). *For the Haar wavelet, the product map* (4) *satisfies $V_{\text{prod}}^\dagger V_{\text{prod}} = I$ for any fusion category $\mathcal{C}$.*

Proof. For Haar, particle p at coarse site x_p spreads to fine sites $\{2x_p-1, 2x_p\}$. Since $x_1 < \cdots < x_N$ are distinct, the fine-site supports $\{2x_p-1, 2x_p\}$ are pairwise disjoint. The constrained sum over $y_1 < \cdots < y_N$ factorises:

$$(V^\dagger V)_{\alpha\beta} = \delta_{\alpha\beta} \prod_{p=1}^{N} \sum_{y_p\in\{2x_p-1,2x_p\}} |R_{y_p,x_p}|^2 = \delta_{\alpha\beta} \prod_{p=1}^{N} 1 = \delta_{\alpha\beta}. \tag{5}$$

The disjointness of supports is the key: it eliminates the hard-core constraint between particles, allowing the sum to factorise. □

1.4 Failure of the product map for overlapping wavelets

For wavelets with wider support (e.g., Daubechies D4 with 4 filter coefficients), adjacent particles at coarse sites i and $i{+}1$ spread to overlapping fine-site ranges: particle i to $\{2i{-}1, 2i, 2i{+}1, 2i{+}2\}$ and particle $i{+}1$ to $\{2i{+}1, 2i{+}2, 2i{+}3, 2i{+}4\}$. The overlap at $\{2i{+}1, 2i{+}2\}$ prevents the sum from factorising, and $V_{\text{prod}}^\dagger V_{\text{prod}} \neq I$.

Remark 1.2. Numerically, the deficit grows with the number of adjacent particle pairs: for D4 on $L = 3$ Fibonacci, $\|V^\dagger V - I\| \approx 0.11$ for the product map.

1.5 The Slater determinant lift for identical particles

For a single non-vacuum simple (e.g., τ in Fibonacci, ψ in sVec), all particles carry the same label. The Hilbert space factorises as

$$\mathcal{H}_N(L) = \Lambda^N(\mathbb{C}^L) \otimes \mathrm{Mor}(X_c, X_a^{\otimes N}) \tag{6}$$

where $\Lambda^N(\mathbb{C}^L) \cong \mathbb{C}^{\binom{L}{N}}$ is the N-th exterior power (the hard-core position space) and $\mathrm{Mor}(X_c, X_a^{\otimes N})$ is the fusion-tree internal space.

Theorem 1.3 (Slater lift for identical anyons). *For a fusion category $\mathcal{C}$ with a single non-vacuum simple object X_a, the fine-graining isometry*

$$V = \Lambda^N(R) \otimes I_{\mathrm{Mor}(X_c, X_a^{\otimes N})} \tag{7}$$

satisfies $V^\dagger V = I$ for any *one-particle isometry $R : \mathbb{C}^L \to \mathbb{C}^{2L}$.*

Proof. The position part $\Lambda^N(R)$ has matrix elements $(\Lambda^N(R))_{S,T} = \det(R_{S,T})$ for N-element subsets $S \subset \{1, \ldots, 2L\}$ and $T \subset \{1, \ldots, L\}$. By Cauchy–Binet:

$$(\Lambda^N(R)^\dagger \Lambda^N(R))_{T,T'} = \sum_{|S|=N} \det(R_{S,T})^* \det\big(R_{S,T'}\big) = \det\Big(R^\dagger R\Big)_{T,T'} = \det(I_L)_{T,T'} = \delta_{TT'}. \quad (8)$$

Since the internal part is tensored with the identity, $V^\dagger V = I \otimes I = I$. □

Remark 1.4. The antisymmetric signs in det are a *mathematical* necessity (for Cauchy–Binet), not a statement about fermionic statistics. The same construction works for Fibonacci (τ anyons with non-abelian braiding), sVec (ψ fermions), or any category with a single particle type.

1.6 The multi-species problem

For categories with multiple non-vacuum simples (e.g., Ising with ψ and σ), configurations can have mixed labels. The factorisation (6) *breaks*: the fusion-tree space $\mathrm{Mor}(X_c, X_{a_1} \otimes \cdots \otimes X_{a_N})$ depends on the specific label sequence $\vec{a}$, which is tied to the positions.

The Slater determinant lift fails for mixed labels because:

1. The determinant introduces cross-terms where particles are permuted.
2. Permuting particles of different types changes the label sequence $\vec{a}$.
3. Different label sequences correspond to different fusion trees (different bra states).
4. The cross-terms contribute to *different* fine states, preventing Cauchy–Binet cancellation.

We investigated several alternative constructions:

- **Categorical determinant** $\sum_{\sigma \in S_N} B(\sigma) \prod_p R_{y_p, x_{\sigma(p)}}$ with braiding weights: reduces the error by $\sim$50% but is not isometric (Cauchy–Binet requires scalar phases, not matrix-valued braiding).
- **Species-enlarged exterior algebra** $\Lambda^N(\mathbb{C}^{Lk})$: the hard-core-on-sites projection (not just hard-core-on-(site, species) pairs) breaks Cauchy–Binet.
- **Label-grouped Slater**: Slater within same-label groups, product across groups. Partial fix only (handles same-species overlaps but not cross-species).

1.7 The normalized product map: the universal construction

Theorem 1.5 (Universal fine-graining isometry)**.** *For any fusion category $\mathcal{C}$ with any number of particle types, and any one-particle isometry R, the* normalized product map

$$V_{\text{iso}} = V_{\text{prod}} \cdot \operatorname{diag}\left(\frac{1}{g(\vec{x})}\right) \quad (9)$$

is isometric ($V_{\text{iso}}^\dagger V_{\text{iso}} = I$), where $g(\vec{x}) = \|V_{\text{prod}} \,|\vec{x}, \vec{a}, \vec{e}, c\rangle\, \|$ is the column norm of the product map.

This relies on two structural properties of the product map, which we have verified numerically for sVec, Fibonacci, and Ising with both Haar and D4 wavelets for $L = 3, 4, 5$:

Claim 1.6 (Diagonal Gram matrix)**.** *The Gram matrix $G = V_{\text{prod}}^\dagger V_{\text{prod}}$ is diagonal: different coarse states map to orthogonal fine-state superpositions.*

Claim 1.7 (Position-only normalization)**.** *The column norm $g(\vec{x}, \vec{a}, \vec{e}, c) = g(\vec{x})$ depends only on the particle positions $\vec{x}$, not on the labels $\vec{a}$, intermediates $\vec{e}$, or total charge c. In particular, it is* category-independent*.*

The normalization factor has a transparent geometric interpretation:

$$g(\vec{x})^2 = \sum_{y_1<\cdots<y_N} \prod_{p=1}^{N} R^2_{y_p,x_p} \tag{10}$$

which is the probability that N independently spread particles all land at distinct sites.

Remark 1.8. For Haar wavelets, $g(\vec{x}) = 1$ for all configurations (non-overlapping supports), recovering the product map. For D4, $g < 1$ when adjacent coarse sites have overlapping fine-site supports. The correction is purely geometric and does not involve any categorical data (F-symbols, R-symbols, etc.).

1.8 Number-changing fine-graining isometry

The normalized product map (9) fixes the deficit $D = I - V^\dagger_{\text{prod}} V_{\text{prod}}$ by rescaling columns, which is mathematically clean but physically artificial. We now show that the deficit can be filled *physically* by pair-creation terms, yielding a number-changing isometry

$$V = V_0 + V_2, \qquad V_0 : \mathcal{H}_N(L) \to \mathcal{H}_N(2L), \quad V_2 : \mathcal{H}_N(L) \to \mathcal{H}_{N+2}(2L) \tag{11}$$

where $V_0 = V_{\text{prod}}$ is the raw product map and V_2 is built from the pair-creation operator $h^\dagger$ [5] applied on the fine lattice.

1.8.1 The deficit

The Gram matrix $G = V_0^\dagger V_0$ is diagonal with entries $g(\vec{x})^2 \leq 1$. The deficit

$$D = I - G = \operatorname{diag}\big(1 - g(\vec{x})^2\big) \tag{12}$$

vanishes for $N = 0, 1$ (no hard-core collisions possible) and grows with the number of adjacent particle pairs. For D4 wavelets, the deficit values are approximately

N	max deficit
0, 1	0
2	0.029
3	0.057
4	0.083

These values are *category-independent*: Fibonacci and Ising produce identical deficit spectra.

1.8.2 Composition with pair creation

The pair-creation operator $h^\dagger$ on the fine lattice maps $\mathcal{H}_N(2L) \to \mathcal{H}_{N+2}(2L)$ by creating vacuum pairs at adjacent vacant sites [5]. The raw composition $h^\dagger V_0$ maps $\mathcal{H}_N(L) \to \mathcal{H}_{N+2}(2L)$, spreading coarse particles via R and then inserting pairs at the newly created vacant fine sites.

The naive construction $V = V_0 + \alpha\, h^\dagger V_0$ fails for two reasons:

1. **Cross-sector overlap**: V_0 also maps coarse $(N{+}2)$-particle states into the fine $(N{+}2)$-particle sector, so $V_0^\dagger (h^\dagger V_0) \neq 0$.
2. **Column interference**: the $h^\dagger V_0$ columns for different coarse states are not orthogonal (different coarse states can produce the same fine state via pair creation at different bonds).

1.8.3 Three-step orthogonalisation

Both obstructions are resolved by a systematic orthogonalisation procedure:

Step 1: Projection. Let $\Pi_{V_0} = V_0 G^{-1} V_0^\dagger$ be the projector onto $\mathrm{Im}(V_0)$, where $G^{-1} = \mathrm{diag}(1/g(\vec{x})^2)$ (well-defined since $g > 0$ for all occupied states). Define

$$V_2^\perp = (I - \Pi_{V_0})\, h^\dagger V_0. \tag{13}$$

This ensures $V_0^\dagger V_2^\perp = 0$ exactly.

Step 2: Löwdin orthogonalisation. Restrict to the columns $J = \{j : D_{jj} > 0\}$ with nonzero deficit. The Gram matrix of the projected columns,

$$G_2 = (V_2^\perp)_J^\dagger (V_2^\perp)_J, \tag{14}$$

is positive definite (verified numerically). Apply symmetric orthogonalisation:

$$(V_2^\perp)_J \longmapsto (V_2^\perp)_J \cdot G_2^{-1/2}. \tag{15}$$

This gives orthonormal V_2 columns within the complement of $\mathrm{Im}(V_0)$.

Step 3: Scaling. Multiply each column $j \in J$ by $\sqrt{D_{jj}}$ to fill the deficit:

$$V_2 = (I - \Pi_{V_0})\, h^\dagger V_0 \cdot G_2^{-1/2} \cdot D_J^{1/2}. \tag{16}$$

Proposition 1.9 (Number-changing isometry)**.** *The map $V = V_0 + V_2$ with V_2 given by* (**??**) *satisfies $V^\dagger V = I$.*

Proof. By construction:

$$V^\dagger V = V_0^\dagger V_0 + \underbrace{V_0^\dagger V_2}_{=0} + \underbrace{V_2^\dagger V_0}_{=0} + V_2^\dagger V_2 = G + D = I. \tag{17}$$

The cross-terms vanish by Step 1. The diagonal term $V_2^\dagger V_2 = D$ by Steps 2–3: the Löwdin orthogonalisation makes the columns orthonormal, and the scaling adjusts each to have squared norm D_{jj}. □

Remark 1.10. The construction requires G_2 to be positive definite on the deficit subspace, i.e., the pair-creation operator must generate enough independent fine-state components orthogonal to $\mathrm{Im}(V_0)$. This has been verified numerically for Fibonacci and Ising at $L = 3, 4$ with D4 wavelets.

1.8.4 Numerical results

We have verified $\|V^\dagger V - I\| < 10^{-14}$ (machine precision) for:

- Fibonacci category, $L = 3, 4$, D4 wavelet
- Ising category (multi-species: σ, ψ), $L = 3, 4$, D4 wavelet

For the Hamiltonian intertwining $V^\dagger H_{2L} V \approx aH_L + bI$, the number-changing isometry gives slightly better residuals than the normalized product map:

	L	Normalized product	Number-changing
Fibonacci	3	0.448	0.434
Fibonacci	4	0.393	0.378

1.8.5 Physical interpretation

The number-changing isometry realises the fine-graining as a two-step physical process:

1. **Spread**: each coarse particle spreads to fine sites via the wavelet filter R (preserving N).
2. **Pair creation**: vacuum pairs are created at vacant fine sites to fill the deficit left by hard-core collisions (changing $N \to N+2$).

The pair-creation contribution is small (scaling factors $\sim \sqrt{D/\|h^\dagger V_0\|^2} \sim 0.1$) but structurally necessary. This provides the first concrete evidence that Garjani–Ardonne pair creation [5] plays a role in the OAR continuum limit [3], not just in the lattice Hamiltonian.

1.9 Hierarchy of constructions

Table 1 summarises which fine-graining constructions are valid in each regime.

Construction	**Single species**	**Multi-species**	**Number-preserving**
Product map (Haar)	✓	✓	yes
Product map (D4)	×	×	yes
Slater $\Lambda^N(R) \otimes I$	✓	×	yes
Normalized product	✓	✓	yes
Number-changing (**??**)	✓	✓	no

Table 1: Fine-graining isometry constructions. The normalized product map (9) and number-changing isometry (**??**) are both universal. The former is simpler; the latter is physically motivated and gives slightly better Hamiltonian intertwining.

1.10 Multi-step composition and the inductive limit

The fine-graining isometries compose correctly: $V_{L\to 4L} = V_{2L\to 4L} \circ V_{L\to 2L}$ is isometric whenever each factor is. This is verified numerically for sVec, Fibonacci, and Ising with D4 wavelets at $L = 3$.

The sequence of lattice algebras $\mathcal{A}_L \xrightarrow{\alpha_{2L,L}} \mathcal{A}_{2L} \xrightarrow{\alpha_{4L,2L}} \mathcal{A}_{4L} \to \cdots$ under the fine-graining $*$-homomorphisms $\alpha_{2L,L}(A) = V^\dagger A V$ forms a directed system whose inductive limit $\mathcal{A}_\infty = \varinjlim \mathcal{A}_N$ is the continuum C*-algebra of observables. The existence and uniqueness of this limit is guaranteed by the general theory of inductive limits of C*-algebras [1].

1.11 Categorical interpretation

The key insight is that the *position* and *internal* degrees of freedom play fundamentally different roles:

- **Position space**: For hard-core particles, the N-body position space is $\binom{L}{N}$-dimensional, isomorphic to $\Lambda^N(\mathbb{C}^L)$ as a vector space (though the isomorphism involves a choice of ordering). The fine-graining map R acts only on positions.
- **Internal space**: The fusion-tree space $\mathrm{Mor}(X_c, X_{a_1} \otimes \cdots \otimes X_{a_N})$ encodes the categorical data. It is independent of positions and is preserved by the fine-graining map.

The categorical data (F-symbols, R-symbols, quantum dimensions) enters the physics through the *Hamiltonian* — hopping, interaction, braiding, and pair creation terms — not through the fine-graining map itself. The fine-graining is a purely geometric operation on the lattice, while the anyonic structure lives in the dynamics.

References

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